7. A great deal of scientific research into global warming has happened over the last 30 years.

More recently, two new contrasting theories linked to global warming have been widely reported. These are outlined below.

Solar activity

A small number of scientists believe that changes in solar activity (changes in the brightness and warmth of the Sun) is causing global warming. Solar activity occurs on the surface of the Sun when nuclear reactions cause flares to be released, high winds to occur and a release of high energy particles from the Sun.

Global dimming

Some scientists believe that global warming is increasing the Earth's mean temperature, but that global dimming is preventing it from increasing even more. Global dimming is the reduction in the amount of radiation reaching the Earth's surface from the Sun. It is thought to be caused by an increase of aerosol particles in the Earth's atmosphere caused by pollution, dust and smog, as well as volcanic eruptions. These particles absorb solar energy and reflect sunlight back into space.

Figure 1 shows the mean atmospheric temperature, carbon dioxide concentration in the atmosphere and solar activity between 1930 and 2010.

Figure 1

Year	Mean atmospheric temperature (°C)	CO_2 concentration (ppm)	Solar activity (arbitrary units)
1930	13.6	309	3.9
1940	13.6	311	4.0
1950	13.8	316	4.2
1960	13.9	320	4.2
1970	14.0	331	4.0
1980	14.2	342	3.9
1990	14.3	358	3.7
2000	14.4	370	3.7
2010	14.6	383	3.6

ppm = parts per million

Examiner only

Figure 2 shows the mean atmospheric temperature within a 20-mile radius of the Mount Pinatubo volcano in the Philippines between 1990 and 2000. Mount Pinatubo erupted in 1991.

Figure 2

Year	Mean atmospheric temperature (°C)
1990	15.1
1991	14.7
1992	14.8
1993	14.7
1994	14.7
1995	14.6
1996	14.7
1997	14.8
1998	14.7
1999	14.8
2000	14.9

(a) Tick (✓) the box that **best** describes the change in carbon dioxide concentration in the atmosphere between 1930 and 2010. [1]

Carbon dioxide concentration increased by approximately 10 ppm every 10 years ☐

Carbon dioxide concentration increased more between 1970 and 2010 than it did between 1930 and 1960 ☐

Carbon dioxide concentration increased more between 1930 and 1960 than it did between 1970 and 2010 ☐

There is no trend to the change in carbon dioxide concentration between 1930 and 2010 ☐

Examiner only

(b) A newspaper report states that global warming is caused by increases in atmospheric carbon dioxide levels **and** by increases in solar activity.

Explain whether the data in **Figure 1** supports this claim. [2]

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(c) **Figure 2** shows that the mean atmospheric temperature near Mount Pinatubo decreased after the volcanic eruption of 1991. Suggest why scientists would **not** use these findings to explain how volcanic activity affects **global** temperatures. [1]

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(d) State **one** industrial method that is being developed to reduce the concentration of carbon dioxide in the atmosphere. [1]

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(e) The Earth's original atmosphere contained about 95% carbon dioxide. Today, the atmosphere contains 0.04% carbon dioxide. Explain why the level of carbon dioxide decreased over geological time. [3]

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(f) When carbon dioxide turns limewater milky, calcium hydroxide is produced.

Give the formula of calcium hydroxide. [1]

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